

TABLE OF CONTENTS

Paragraph Number	Title	Page Number
Section 1		
Introduction		
1.1	Features	1-2
1.2	Programming Model	1-4
1.3	Data Types and Addressing Modes Overview	1-8
1.4	Instruction Set Overview	1-10
1.5	Virtual Memory and Virtual Machine Concepts	1-10
1.5.1	Virtual Memory	1-10
1.5.2	Virtual Machine	1-12
1.6	Pipelined Architecture	1-12
1.7	Cache Memory	1-13
Section 2		
Processing States		
2.1	Privilege Levels	2-2
2.1.1	Supervisor Privilege Level	2-2
2.1.2	User Privilege Level	2-3
2.1.3	Changing Privilege Level	2-3
2.2	Address Space Types	2-4
2.3	Exception Processing	2-5
2.3.1	Exception Vectors	2-5
2.3.2	Exception Stack Frame	2-6
Section 3		
Signal Description		
3.1	Signal Index	3-2
3.2	Function Code Signals (FC2–FC0)	3-2
3.3	Address Bus (A31–A0, MC68020)(A23–A0, MC68EC020)	3-2
3.4	Data Bus (D31–D0)	3-2
3.5	Transfer Size Signals (SIZ1, SIZ0)	3-2
3.6	Asynchronous Bus Control Signals	3-4
3.7	Interrupt Control Signals	3-5
3.8	Bus Arbitration Control Signals	3-6
3.9	Bus Exception Control Signals	3-6
3.10	Emulator Support Signal	3-7
3.11	Clock (CLK)	3-7

TABLE OF CONTENTS (Continued)

Paragraph Number	Title	Page Number
3.12	Power Supply Connections	3-7
3.13	Signal Summary	3-8
Section 4		
On-Chip Cache Memory		
4.1	On-Chip Cache Organization and Operation	4-1
4.2	Cache Reset	4-3
4.3	Cache Control	4-3
4.3.1	Cache Control Register (CACR)	4-3
4.3.2	Cache Address Register (CAAR)	4-4
Section 5		
Bus Operation		
5.1	Bus Transfer Signals	5-1
5.1.1	Bus Control Signals	5-2
5.1.2	Address Bus	5-3
5.1.3	Address Strobe	5-3
5.1.4	Data Bus	5-3
5.1.5	Data Strobe	5-4
5.1.6	Data Buffer Enable	5-4
5.1.7	Bus Cycle Termination Signals	5-4
5.2	Data Transfer Mechanism	5-5
5.2.1	Dynamic Bus Sizing	5-5
5.2.2	Misaligned Operands	5-14
5.2.3	Effects of Dynamic Bus Sizing and Operand Misalignment	5-20
5.2.4	Address, Size, and Data Bus Relationships	5-21
5.2.5	Cache Interactions	5-22
5.2.6	Bus Operation	5-24
5.2.7	Synchronous Operation with DSACK1/DSACK0	5-24
5.3	Data Transfer Cycles	5-25
5.3.1	Read Cycle	5-26
5.3.2	Write Cycle	5-33
5.3.3	Read-Modify-Write Cycle	5-39
5.4	CPU Space Cycles	5-44
5.4.1	Interrupt Acknowledge Bus Cycles	5-45
5.4.1.1	Interrupt Acknowledge Cycle—Terminated Normally	5-45
5.4.1.2	Autovector Interrupt Acknowledge Cycle	5-48
5.4.1.3	Spurious Interrupt Cycle	5-48
5.4.2	Breakpoint Acknowledge Cycle	5-50
5.4.3	Coprocessor Communication Cycles	5-53
5.5	Bus Exception Control Cycles	5-53
5.5.1	Bus Errors	5-55

TABLE OF CONTENTS (Continued)

Paragraph Number	Title	Page Number
5.5.2	Retry Operation	5-56
5.5.3	Halt Operation	5-60
5.5.4	Double Bus Fault	5-60
5.6	Bus Synchronization	5-62
5.7	Bus Arbitration	5-62
5.7.1	MC68020 Bus Arbitration	5-63
5.7.1.1	Bus Request (MC68020)	5-66
5.7.1.2	Bus Grant (MC68020)	5-66
5.7.1.3	Bus Grant Acknowledge (MC68020)	5-66
5.7.1.4	Bus Arbitration Control (MC68020)	5-67
5.7.2	MC68EC020 Bus Arbitration	5-70
5.7.2.1	Bus Request (MC68EC020)	5-71
5.7.2.2	Bus Grant (MC68EC020)	5-71
5.7.2.3	Bus Arbitration Control (MC68EC020)	5-73
5.8	Reset Operation	5-76

Section 6 Exception Processing

6.1	Exception Processing Sequence	6-1
6.1.1	Reset Exception	6-4
6.1.2	Bus Error Exception	6-4
6.1.3	Address Error Exception	6-6
6.1.4	Instruction Trap Exception	6-6
6.1.5	Illegal Instruction and Unimplemented Instruction Exceptions	6-7
6.1.6	Privilege Violation Exception	6-8
6.1.7	Trace Exception	6-9
6.1.8	Format Error Exception	6-10
6.1.9	Interrupt Exceptions	6-11
6.1.10	Breakpoint Instruction Exception	6-17
6.1.11	Multiple Exceptions	6-17
6.1.12	Return from Exception	6-19
6.2	Bus Fault Recovery	6-21
6.2.1	Special Status Word (SSW)	6-21
6.2.2	Using Software to Complete the Bus Cycles	6-23
6.2.3	Completing the Bus Cycles with RTE	6-24
6.3	Coprocessor Considerations	6-25
6.4	Exception Stack Frame Formats	6-25

TABLE OF CONTENTS (Continued)

Paragraph Number	Title	Page Number
Section 7		
Coprocessor Interface Description		
7.1	Introduction	7-1
7.1.1	Interface Features	7-2
7.1.2	Concurrent Operation Support	7-2
7.1.3	Coprocessor Instruction Format	7-3
7.1.4	Coprocessor System Interface	7-4
7.1.4.1	Coprocessor Classification	7-4
7.1.4.2	Processor-Coprocessor Interface	7-5
7.1.4.3	Coprocessor Interface Register Selection	7-6
7.2	Coprocessor Instruction Types	7-7
7.2.1	Coprocessor General Instructions	7-8
7.2.1.1	Format	7-8
7.2.1.2	Protocol	7-9
7.2.2	Coprocessor Conditional Instructions	7-10
7.2.2.1	Branch on Coprocessor Condition Instruction	7-12
7.2.2.1.1	Format	7-12
7.2.2.1.2	Protocol	7-12
7.2.2.2	Set on Coprocessor Condition Instruction	7-13
7.2.2.2.1	Format	7-13
7.2.2.2.2	Protocol	7-14
7.2.2.3	Test Coprocessor Condition, Decrement, and Branch Instruction ...	7-14
7.2.2.3.1	Format	7-14
7.2.2.3.2	Protocol	7-15
7.2.2.4	Trap on Coprocessor Condition Instruction	7-15
7.2.2.4.1	Format	7-15
7.2.2.4.2	Protocol	7-16
7.2.3	Coprocessor Context Save and Restore Instructions	7-16
7.2.3.1	Coprocessor Internal State Frames	7-17
7.2.3.2	Coprocessor Format Words	7-18
7.2.3.2.1	Empty/Reset Format Word	7-18
7.2.3.2.2	Not-Ready Format Word	7-19
7.2.3.2.3	Invalid Format Word	7-19
7.2.3.2.4	Valid Format Word	7-20
7.2.3.3	Coprocessor Context Save Instruction	7-20
7.2.3.3.1	Format	7-20
7.2.3.3.2	Protocol	7-21
7.2.3.4	Coprocessor Context Restore Instruction	7-22
7.2.3.4.1	Format	7-22
7.2.3.4.2	Protocol	7-23
7.3	Coprocessor Interface Register Set	7-24

TABLE OF CONTENTS (Continued)

Paragraph Number	Title	Page Number
7.3.1	Response CIR	7-24
7.3.2	Control CIR	7-24
7.3.3	Save CIR	7-25
7.3.4	Restore CIR	7-25
7.3.5	Operation Word CIR	7-25
7.3.6	Command CIR	7-25
7.3.7	Condition CIR	7-26
7.3.8	Operand CIR	7-26
7.3.9	Register Select CIR	7-27
7.3.10	Instruction Address CIR	7-27
7.3.11	Operand Address CIR	7-27
7.4	Coprocessor Response Primitives	7-27
7.4.1	ScanPC	7-28
7.4.2	Coprocessor Response Primitive General Format	7-28
7.4.3	Busy Primitive	7-30
7.4.4	Null Primitive	7-31
7.4.5	Supervisor Check Primitive	7-33
7.4.6	Transfer Operation Word Primitive	7-33
7.4.7	Transfer from Instruction Stream Primitive	7-34
7.4.8	Evaluate and Transfer Effective Address Primitive	7-35
7.4.9	Evaluate Effective Address and Transfer Data Primitive	7-35
7.4.10	Write to Previously Evaluated Effective Address Primitive	7-37
7.4.11	Take Address and Transfer Data Primitive	7-39
7.4.12	Transfer to/from Top of Stack Primitive	7-40
7.4.13	Transfer Single Main Processor Register Primitive	7-40
7.4.14	Transfer Main Processor Control Register Primitive	7-41
7.4.15	Transfer Multiple Main Processor Registers Primitive	7-42
7.4.16	Transfer Multiple Coprocessor Registers Primitive	7-42
7.4.17	Transfer Status Register and ScanPC Primitive	7-44
7.4.18	Take Preinstruction Exception Primitive	7-45
7.4.19	Take Midinstruction Exception Primitive	7-47
7.4.20	Take Postinstruction Exception Primitive	7-48
7.5	Exceptions	7-49
7.5.1	Coprocessor-Detected Exceptions	7-49
7.5.1.1	Coprocessor-Detected Protocol Violations	7-50
7.5.1.2	Coprocessor-Detected Illegal Command or Condition Words	7-51
7.5.1.3	Coprocessor Data-Processing-Related Exceptions	7-51
7.5.1.4	Coprocessor System-Related Exceptions	7-51
7.5.1.5	Format Errors	7-52
7.5.2	Main-Processor-Detected Exceptions	7-52
7.5.2.1	Protocol Violations	7-52
7.5.2.2	F-Line Emulator Exceptions	7-54

TABLE OF CONTENTS (Continued)

Paragraph Number	Title	Page Number
7.5.2.3	Privilege Violations.....	7-55
7.5.2.4	cpTRAPcc Instruction Traps	7-55
7.5.2.5	Trace Exceptions	7-55
7.5.2.6	Interrupts	7-56
7.5.2.7	Format Errors	7-57
7.5.2.8	Address and Bus Errors.....	7-57
7.5.3	Coprocessor Reset	7-58
7.6	Coprocessor Summary	7-58

Section 8 Instruction Execution Timing

8.1	Timing Estimation Factors	8-1
8.1.1	Instruction Cache and Prefetch	8-1
8.1.2	Operand Misalignment	8-2
8.1.3	Bus/Sequencer Concurrency.....	8-2
8.1.4	Instruction Execution Overlap	8-3
8.1.5	Instruction Stream Timing Examples	8-4
8.2	Instruction Timing Tables	8-9
8.2.1	Fetch Effective Address	8-13
8.2.2	Fetch Immediate Effective Address.....	8-14
8.2.3	Calculate Effective Address	8-16
8.2.4	Calculate Immediate Effective Address.....	8-17
8.2.5	Jump Effective Address.....	8-19
8.2.6	MOVE Instruction	8-20
8.2.7	Special-Purpose MOVE Instruction	8-29
8.2.8	Arithmetic/Logical Instructions.....	8-30
8.2.9	Immediate Arithmetic/Logical Instructions	8-31
8.2.10	Binary-Coded Decimal Operations	8-32
8.2.11	Single-Operand Instructions	8-33
8.2.12	Shift/Rotate Instructions	8-34
8.2.13	Bit Manipulation Instructions	8-35
8.2.14	Bit Field Manipulation Instructions.....	8-36
8.2.15	Conditional Branch Instructions.....	8-37
8.2.16	Control Instructions.....	8-38
8.2.17	Exception-Related Instructions	8-39
8.2.18	Save and Restore Operations	8-40

Section 9 Applications Information

9.1	Floating-Point Units	9-1
9.2	Byte Select Logic for the MC68020/EC020	9-5
9.3	Power and Ground Considerations	9-9

TABLE OF CONTENTS (Concluded)

Paragraph Number	Title	Page Number
9.4	Clock Driver	9-10
9.5	Memory Interface	9-11
9.6	Access Time Calculations	9-12
9.7	Module Support	9-14
9.7.1	Module Descriptor	9-14
9.7.2	Module Stack Frame	9-16
9.8	Access Levels	9-17
9.8.1	Module Call	9-18
9.8.2	Module Return	9-19

Section 10 Electrical Characteristics

10.1	Maximum Ratings	10-1
10.2	Thermal Considerations	10-1
10.2.1	MC68020 Thermal Characteristics and DC Electrical Characteristics	10-2
10.2.2	MC68EC020 Thermal Characteristics and DC Electrical Characteristics	10-4
10.3	AC Electrical Characteristics	10-5

Section 11 Ordering Information and Mechanical Data

11.1	Standard Ordering Information	11-1
11.1.1	Standard MC68020 Ordering Information	11-1
11.1.2	Standard MC68EC020 Ordering Information	11-1
11.2	Pin Assignments and Package Dimensions	11-2
11.2.1	MC68020 RC and RP Suffix—Pin Assignment	11-2
11.2.2	MC68020 RC Suffix—Package Dimensions	11-3
11.2.3	MC68020 RP Suffix—Package Dimensions	11-4
11.2.4	MC68020 FC and FE Suffix—Pin Assignment	11-5
11.2.5	MC68020 FC Suffix—Package Dimensions	11-6
11.2.6	MC68020 FE Suffix—Package Dimensions	11-7
11.2.7	MC68EC020 RP Suffix—Pin Assignment	11-8
11.2.8	MC68EC020 RP Suffix—Package Dimensions	11-9
11.2.9	MC68EC020 FG Suffix—Pin Assignment	11-10
11.2.10	MC68EC020 FG Suffix—Package Dimensions	11-11

Appendix A Interfacing an MC68EC020 to a DMA Device That Supports a Three-Wire Bus Arbitration Protocol